

Topology General Exam

January 9, 2019

Instructions: This is a four hour exam. Your solutions should be legible and clearly organized, written in complete sentences in good mathematical style. All work should be your own - no outside sources are permitted - using methods and results from the first year topology course topics.

Problem 1

Let $f : M \rightarrow N$ be a smooth map between manifolds of dimension m and n , respectively. Let $q \in N$ be a regular value for f , and let $X = f^{-1}(q) \subset M$.

- a) Prove that if M is orientable, then X is orientable.
- b) Let $g : K \rightarrow M$ be another smooth map, where K is a smooth manifold. Prove that $q \in N$ is a regular value for $f \circ g$ if and only if g is transverse to X .

Problem 2

- a) Prove that any smooth map $f : S^k \rightarrow \mathbb{R}^n$ can be extended to a smooth map $F : D^{k+1} \rightarrow \mathbb{R}^n$, where $S^k = \partial D^{k+1}$ is the k -dimensional sphere and D^{k+1} the unit ball of dimension $k + 1$.
- b) Let $M \subset \mathbb{R}^n$ be a smooth compact manifold of dimension m , and assume $k < n - m - 1$. Prove that any smooth map $f : S^k \rightarrow \mathbb{R}^n - M$ can be extended to a smooth map $F : D^{k+1} \rightarrow \mathbb{R}^n - M$.

Problem 3

Let M and N be the subsets of $\mathbb{R}^3$ defined by

$$M = \{x^2 + y^2 + z^2 = 1\} \quad N = \{x^2 - y^2 + z^2 = c\}$$

for a real number c . Justify your responses to the following:

- a) Determine all values of c for which M and N are submanifolds of $\mathbb{R}^3$, and the intersection $M \cap N$ is transverse.
- b) Determine all values of c for which $M \cap N$ is a submanifold of $\mathbb{R}^3$.

Problem 4

Let X and Y be closed, compact, oriented manifolds of the same dimension. Let $f, g : X \rightarrow Y$ be two smooth maps. The graphs of f and g are the submanifolds of $X \times Y$ given by

$$\Gamma_f = \{(x, f(x)) \mid x \in X\} \quad \Gamma_g = \{(x, g(x)) \mid x \in X\}$$

oriented so that the diffeomorphisms $X \rightarrow \Gamma_f$ and $X \rightarrow \Gamma_g$ given by $x \mapsto (x, f(x))$ and $x \mapsto (x, g(x))$ are orientation-preserving.

The coincidence number of f and g , written $C(f, g)$, is defined to be the intersection number $I(\Gamma_f, \Gamma_g) \in \mathbb{Z}$ (sometimes also written $\Gamma_f \cdot \Gamma_g$).

a) Prove that if $C(f, g) \neq 0$ then for any smooth maps $\tilde{f}, \tilde{g} : X \rightarrow Y$ such that $\tilde{f}$ and $\tilde{g}$ are homotopic to f and g , respectively, there exists a point $x \in X$ such that $\tilde{f}(x) = \tilde{g}(x)$.

b) Let $f, g : S^1 \rightarrow S^1$ be two smooth maps of degree n and m , respectively. Prove that if $n \neq m$, then there is a point $x \in S^1$ with $f(x) \neq g(x)$.

Problem 5

Let $p : S^n \rightarrow \mathbb{R}P^n$ be the projection.

a) Let $f : \mathbb{R}P^n \rightarrow \mathbb{R}P^n$ be continuous. Show that then there exists a continuous map $\tilde{f} : S^n \rightarrow S^n$ such that $p \circ \tilde{f} = f \circ p : S^n \rightarrow \mathbb{R}P^n$.

b) Show that every continuous map $f : \mathbb{R}P^{2k} \rightarrow \mathbb{R}P^{2k}$ has a fixed point.

Problem 6

Suppose given a commutative diagram of abelian groups with exact rows:

$$\begin{array}{ccccccccc} C_1 & \xrightarrow{\alpha_1} & C_2 & \xrightarrow{\alpha_2} & C_3 & \xrightarrow{\alpha_3} & C_4 & \xrightarrow{\alpha_4} & C_5 \\ \downarrow f_1 & & \downarrow f_2 & & \downarrow f_3 & & \downarrow f_4 & & \downarrow f_5 \\ D_1 & \xrightarrow{\beta_1} & D_2 & \xrightarrow{\beta_2} & D_3 & \xrightarrow{\beta_3} & D_4 & \xrightarrow{\beta_4} & D_5 \end{array}$$

The top row is $C_1 \rightarrow C_2 \rightarrow C_3 \rightarrow C_4 \rightarrow C_5$, with consecutive maps $\alpha_1, \alpha_2, \alpha_3, \alpha_4$. The bottom row is $D_1 \rightarrow D_2 \rightarrow D_3 \rightarrow D_4 \rightarrow D_5$, with consecutive maps $\beta_1, \beta_2, \beta_3, \beta_4$. For each r from 1 to 5, the vertical arrow f_r maps C_r downward to D_r . Each square commutes and both rows are exact.

Prove part of the Five Lemma: show that if f_2 and f_4 are monomorphisms, and f_1 is an epimorphism, then f_3 is a monomorphism. (Hint: start by showing that $x \in \text{Ker}(f_3) \Rightarrow x \in \text{Ker}(\alpha_3)$.)

Problem 7

Let $C \subset \mathbb{R}^3$ be the union of the x -axis and the y -axis. Compute $H^*(\mathbb{R}^3 - C; \mathbb{Z})$. (Hint: note that $\mathbb{R}^3 - C = (\mathbb{R}^3 - x\text{-axis}) \cap (\mathbb{R}^3 - y\text{-axis})$.)

Problem 8

Let $a : S^1 \rightarrow S^1 \vee S^1$ and $b : S^1 \rightarrow S^1 \vee S^1$ respectively be the inclusion of the circle as the first and second wedge summand. Then $\pi_1(S^1 \vee S^1)$ can be identified with the free group on a and b . Let $f : S^1 \rightarrow S^1$ represent the element $c \in \pi_1(S^1 \vee S^1)$, and let $X_f = (S^1 \vee S^1) \cup_f D^2$, the topological space obtained by identifying points on $S^1 = \partial D^2$ with their images under f in $S^1 \vee S^1$. If $c = a(ab)^4a$, give a presentation of $\pi_1(X_f)$ and compute $H^*(X_f; \mathbb{Z})$, describing the homology groups as direct sums of cyclic groups, as usual.